

Washu2-CPU Week-7 Report

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Bug Report:

- Investigated Quartus RAM inference issue where the 4096×16 initialized RAM was synthesized as registers instead of embedded memory.
- Debugged RAM implementation by testing different initialization methods and identifying that inline initialization prevents block RAM inference.
- Verified the CPU fetch cycle and confirmed correct operation of the PC, MAR, MBR, IR, and FSM state transitions through simulation.
- Identified undefined (XXXX) values in the Datapath during CLOAD execution, indicating an issue in the immediate/sign-extension path.
- Explored different approaches for program loading into RAM, including runtime initialization, testbench initialization, and external memory initialization, for future implementation.



